Robustness of AI-Text Detection Under Text Editing and Paraphrasing

AI26501 Modern AI Assignments • SP-2026 • Assignment 16 • Group 16

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Presenter/team names: complete before submission

Abstract

This study evaluates a TF–IDF word-unigram/bigram representation with a LinearSVC detector trained on original passages from the HC3 Human–ChatGPT Comparison Corpus. The 300-topic held-out paired evaluation compares original AI text, a legacy phrase-removal edit that changed no sampled passages, a verified deterministic light-edit condition, model-generated T5 paraphrases, and a truncation control matched to paraphrase length. AI recall was 98.00% for original text, 97.00% after the verified light edits, 79.67% for T5 paraphrases, and 89.33% for truncation. The paraphrases had a mean length ratio of 0.561 relative to their originals, and the truncation control also reduced recall. Thus, the raw paraphrase result reflects both rewriting and substantial compression; it cannot be interpreted as the isolated causal effect of meaning-preserving paraphrase.

1. Problem statement and research question

AI-generated passages may be edited or rewritten before publication. A detector can perform well on untouched text but be sensitive to wording, length, or other dataset-specific patterns. The research question is: how does a classical detector's AI recall change under controlled lexical edits, generated paraphrases, and a length-matched truncation control? Primary outcome: AI recall. Secondary outcomes: accuracy, AI precision, AI F1, false-positive rate, confusion-matrix counts, and paired prediction changes.

2. Dataset and split

The study uses a prepared English subset of Hello-SimpleAI/HC3, with finance, medicine, open_qa, reddit_eli5, and wiki_csai sources. The grouped split uses GroupShuffleSplit(test_size=0.20, random_state=42) grouped by topic_id. The saved split has 60,955 training passages, 15,157 test passages, 17,528 training topics, 4,383 held-out topics, and zero topic overlap. The paired pilot contains 300 held-out topics, each with one human control and one AI passage per condition. Source composition is imbalanced: Reddit ELI5 contributes 222 passages, Finance 38, Open QA 22, Medicine 15, and Wiki CSAI 3.

3. Methods and controls

The detector uses TF–IDF word unigrams and bigrams (min_df=2, max_features=200,000, sublinear_tf=True, strip_accents='unicode') and LinearSVC(class_weight='balanced', random_state=42). It is trained only on original training text. For each condition, the same 300 human controls are combined with 300 AI passages, giving 600 passages per condition.

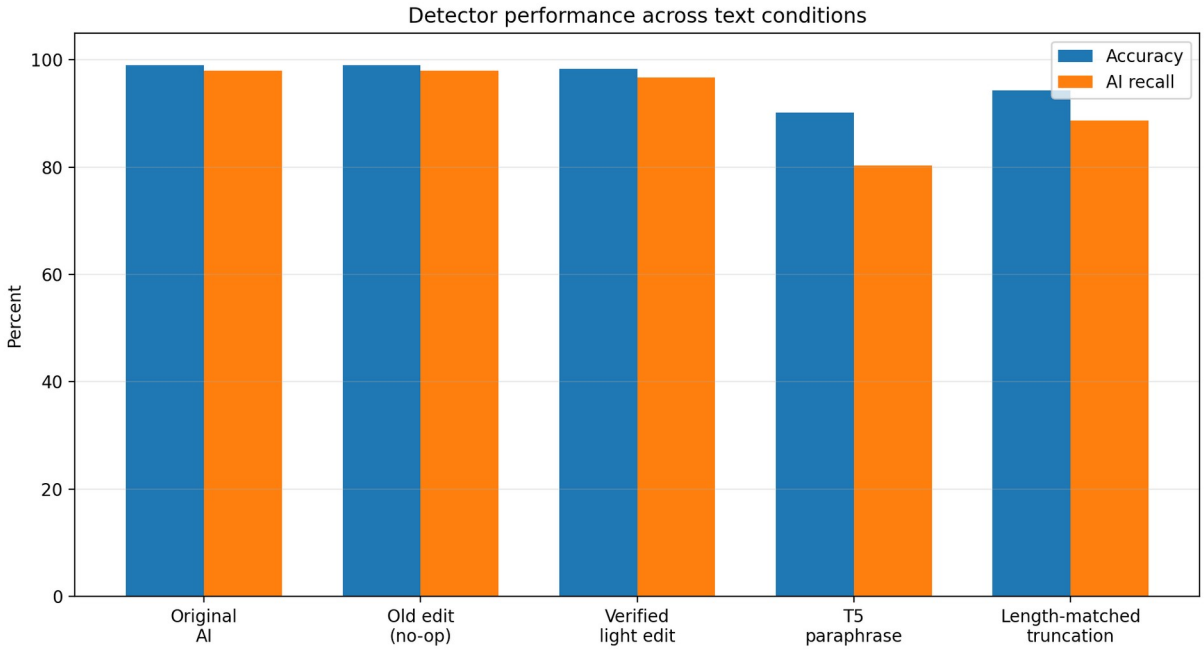
- Original AI: original AI passage for each selected topic.
- Legacy surface edit: the existing opening-phrase-removal function. It changed 0/300 passages in this sample and is retained as a no-op diagnostic, not evidence of edit robustness.
- Verified light edit: a deterministic rule-based lexical edit designed to change text where listed patterns occur. It changed 104/300 paired AI passages; mean length ratio was 0.999 relative to original text.
- T5 paraphrase: generated by humarin/chatgpt_paraphraser_on_T5_base. These are raw model outputs and are not guaranteed to preserve meaning.
- Truncation control: the original passage is truncated to the paraphrase's whitespace-word count when the original is longer. It changed 299/300 passages. For topic 11934, the original had 31 words and the paraphrase 33, so the original was left unchanged. This is a compression control, not a semantic-preservation control; truncation may remove claims or context.

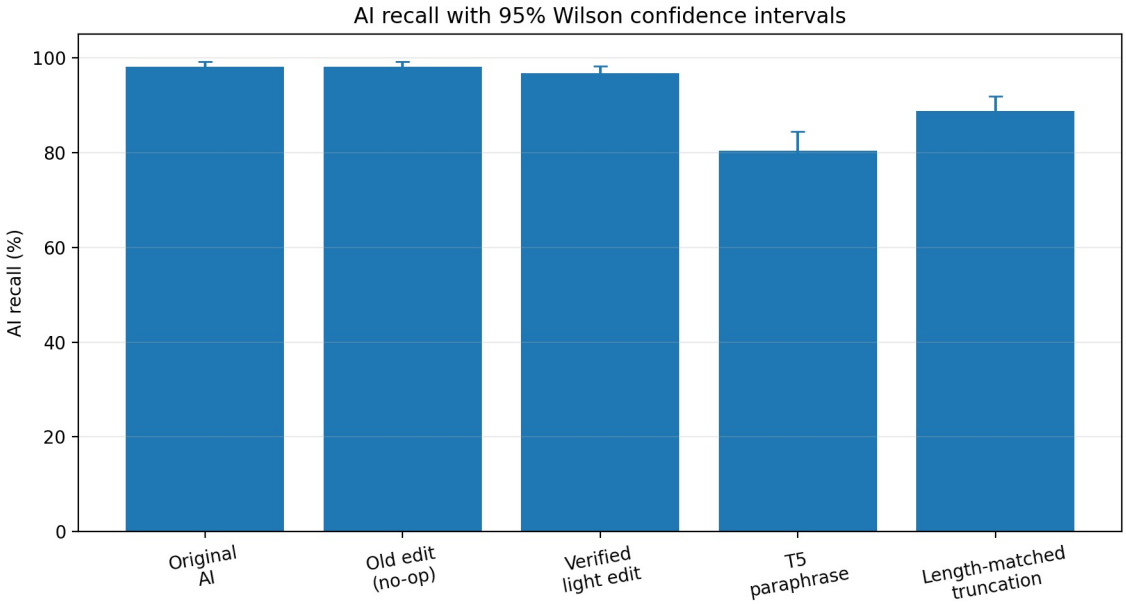
Uncertainty: recall confidence intervals are 95% Wilson intervals. Paired changes in binary AI detections are assessed with exact McNemar tests. These tests compare paired predictions and do not address paraphrase fidelity or all sources of selection bias.

4. Results

Condition	Accuracy	AI precision	AI recall (95% CI)	AI F1	FPR	FN / TP
Original AI	98.83%	99.66%	98.00% (95.71–99.08)	98.82%	0.33%	6 / 294
Legacy phrase-removal	98.83%	99.66%	98.00% (95.71–99.08)	98.82%	0.33%	6 / 294
Verified light edit	98.33%	99.66%	97.00% (94.40–98.41)	98.31%	0.33%	9 / 291
T5 paraphrase	89.67%	99.58%	79.67% (74.75–83.83)	88.52%	0.33%	61 / 239
Truncation control	94.50%	99.63%	89.33% (85.33–92.34)	94.20%	0.33%	32 / 268

Each condition contains 300 human passages and 300 AI passages. In every condition, 299/300 human passages were classified as human and one was incorrectly flagged as AI (FPR = 0.33%). This observed rate is based on only 300 human controls.





5. Paired tests and interpretation

Comparison	AI detected → missed	AI missed → detected	Exact McNemar p
Original vs legacy phrase-removal	0	0	1.000
Original vs verified light edit	3	0	0.250
Original vs T5 paraphrase	57	2	6.14e-15
Original vs truncation control	26	0	2.98e-08
Truncation control vs	38	9	2.49e-05

T5 paraphrase

The legacy phrase-removal edit changed no passages and no predictions. The verified light edit changed 104/300 passages; recall decreased from 98.00% to 97.00% (3 detected-to-missed, 0 missed-to-detected; exact p=0.25), which is not statistically significant at 0.05. Recall decreased to 79.67% for T5 paraphrases (57 detected-to-missed, 2 missed-to-detected; $p \approx 6.14 \times 10^{-15}$). The truncation control achieved 89.33% recall (26 detected-to-missed, 0 missed-to-detected; $p \approx 2.98 \times 10^{-8}$). Mean word-length ratios were 0.561 for paraphrases and 0.561 for truncation. T5 paraphrases had lower recall than the truncation control (38 detected-to-missed, 9 missed-to-detected; paired $p \approx 2.49 \times 10^{-5}$), but this comparison does not isolate a pure semantic effect because paraphrases may also alter content and fidelity.

6. Larger-sample supplementary light-edit experiment

Across all 4,959 held-out AI passages, the detector flagged 4,878 original passages as AI (98.37%) and 4,861 deterministically edited passages as AI (98.02%). The paired predictions had 17 changed classifications (the paired detection-rate decrease was 0.34 percentage points). The human-only held-out group had 75/10,198 false positives (0.74%). This larger experiment uses a separate deterministic edit procedure and should not be conflated with the 300-topic paired pilot.

7. Limitations

- Generated paraphrases were not independently reviewed and rated for meaning preservation. Some outputs omit details or alter factual content.
- The T5 paraphrases are substantially shorter: mean length ratio 0.561; 181/300 were under 60% of their original word count. The truncation control demonstrates that compression alone is associated with lower recall, although it does not fully explain the paraphrase result. One original was shorter than its paraphrase, so the truncation control has one word-count mismatch.
- The paired sample is small and source-imbalanced, dominated by Reddit ELI5. Source-specific findings, especially Wiki CSAI (n=3), are highly uncertain.
- Only one dataset, one principal classical detector, and one paraphrasing model were evaluated.
- Results are conditional on the chosen dataset, split, sampling rules, transformation rules, and model implementation. Detector predictions are not proof of authorship.

8. Conclusion and future work

The verified light-edit condition caused a small, statistically inconclusive recall change in the paired sample, while generated T5 paraphrases produced a larger decrease. The length-matched truncation control also reduced recall, showing that the paraphrase result is partly entangled with text compression. The defensible conclusion is that this detector's performance varies under these specific text transformations; the experiment does not establish universal robustness or isolate meaning-preserving paraphrasing.

Next steps: human-review paraphrase fidelity; run a quality-filtered analysis; expand and balance the sample; evaluate more detectors and paraphrasers; and test human edits and multiple split seeds.

References

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